Artem Matveev

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About Me

Research scientist and engineer with 4+ years of experience in recommender systems, with accepted papers at KDD, SIGIR, and RecSys. Currently a Senior Research Scientist at AI VK. B.Sc. from Lomonosov Moscow State University; M.Sc. candidate at the Higher School of Economics.

Research Interests

Information retrieval, conversational recommendation, large language models, generative recommendation, in-text reasoning for recommendation, reinforcement learning, and the mathematical foundations of deep learning.

Work Experience

Mar 2026 – Present **Senior Research Scientist, AI VK**

Research team led by Prof. Dr. Alexander D'yakonov, formerly world's #1 rank on Kaggle.

- Worked on long-term RL optimization for generative retrieval in recommender systems; first author of *Long-Term Optimization for Large-Scale Generative Retrieval with Off-Policy REINFORCE* (KDD 2026).
- Made major contributions to the internal research framework, speeding up experiment iteration several-fold.

Jul 2025 – Mar 2026 **RecSys R&D Team Lead / Senior Research Engineer, Yandex**

- Led a three-person R&D team responsible for pretraining generative recommendation models, internal training framework development, and inference optimization.
- Worked on pretraining a OneRec-style generative recommender that replaces the entire cascade pipeline (candidate generation, ranking, and re-ranking) with a single end-to-end neural network.
- Introduced FSDP (Fully Sharded Data Parallel) training, scaling recommender transformers up to 7 billion parameters with improved model quality.
- Built the training framework and infrastructure for large-scale generative recommender models.
- Worked on adapting sub-quadratic attention mechanisms for generative retrieval in recommender systems (paper).

May 2023 – Jul 2025 **RecSys Research Engineer, Yandex**

- Investigated graph neural networks for item representation in recommender systems (talk).
- Developed a transformer-based candidate generation model for Yandex Market and conducted A/B tests.
- Co-developed Argus, a billion-parameter generative recommender transformer; achieved +2.26% Total Listening Time and +6.37% item like rate at Yandex Music in online A/B tests; later adapted across Yandex products (e.g., e-commerce +1.7% GMV, food-tech +2.1% GMV).
- Contributed to Argus adaptation for Yandex Ads ranking, delivering +2% revenue uplift in online A/B tests (talk).
- Mentored two interns, who have since become a Senior RecSys Researcher and a core developer at YandexGPT.

Feb 2022 – Jul 2022 **Backend Developer Intern, Yandex**

- C++ backend development: optimized runtime performance of search algorithms and data-preparation pipelines.
- Implemented new ML features and conducted A/B tests.

Education

Sep 2024 – Present **Higher School of Economics (HSE)**, Moscow, Russia

Master's Program in Modern Computer Science.

Thesis: *Long-Term Optimization for Large-Scale Generative Retrieval.*

Sep 2024 – Present **Yandex School of Data Analysis (YSDA)**, Moscow, Russia

Selective postgraduate program in machine learning, deep learning, and applied mathematics.

Sep 2020 – Jun 2024 **Lomonosov Moscow State University (MSU)**, Faculty of CMC, Moscow, Russia

B.Sc. in Applied Mathematics and Computer Science.

Thesis: *Graph Neural Networks for Candidate Retrieval in E-Commerce.*

Selected Publications

KDD 2026 — Long-Term Optimization for Large-Scale Generative Retrieval with Off-Policy REINFORCE, *5th Workshop on End-to-End Customer Journey Optimization*
Artem Matveev, Sergei Makeev, Aleksei Krasilnikov, Vladimir Baikalov, Sergei Liamaev, Kirill Khrylchenko

Scaling Recommender Transformers to One Billion Parameters
Kirill Khrylchenko, **Artem Matveev**, Sergei Makeev, Vladimir Baikalov

SIGIR 2026 — Gated Bidirectional Linear Attention for Generative Retrieval
Artem Matveev, Vladislav Tytskiy, Sergei Makeev, Sergei Liamaev

RecSys 2025 — Correcting the LogQ Correction: Revisiting Sampled Softmax for Large-Scale Retrieval
Kirill Khrylchenko, Vladimir Baikalov, Sergei Makeev, **Artem Matveev**, Sergei Liamaev

arXiv preprint — Embedding Items at Scale: Comparing GNN-Based and ID-Based Item Embeddings in the Yandex Ecosystem
Sergei Makeev, **Artem Matveev**, Vladimir Baikalov, Kirill Khrylchenko

Teaching

2024 – 2026 — Teaching Assistant, Yandex School of Data Analysis (YSDA)
DeepRecSys Course. GitHub (140+ stars).
- Seminars on evaluation metrics, neural RecSys framework design, neural ranking, and reinforcement learning.
- Authored homework assignments for the course.

2025 – 2026 — Teaching Assistant, Higher School of Economics
Recommender Systems Course. GitHub (90+ stars).

2025 — Teaching Assistant, Higher School of Economics
Deep Learning Course. GitHub (100+ stars).
- Seminar on neural ranking.

Talks

2025 — *Do We Need Graph Neural Networks to Encode Items in Recommendations?*, Yandex For ML
YouTube video.

Competitions

2023 — Kaggle "ICR — Identifying Age-Related Conditions"
Silver Medalist.

2022 — ACM ICPC, 2022 ICPC Northern Eurasia Finals
Diploma of the 3rd degree.

2020 — ACM ICPC, Final of Northern Eurasia 2020
Diploma of the 2nd degree.

Technical Skills

Languages	Current: Python, Triton, CUDA. Past: C++, Assembly.
Frameworks	PyTorch.
ML Engineering	Full production cycle: data preparation, experimentation, model export & conversion, performance benchmarking, Triton Inference Server deployment, real-time history processing.

Professional Service

2026 — UMAP 2026 — PC member, Industry Track

Patents

Dec 2025 — Application for invention 2025136959, Federal Service for Intellectual Property
Yandex filed a patent application for Argus (see *Scaling Recommender Transformers to One Billion Parameters*).